

From Computational Metaphysics towards Computational (pseudo) Ethics

Christoph Benz Müller

University of Luxembourg | Freie Universität Berlin | Saarland University

`$HOME/GITHUBS/DeonticCorner/presentation/2018-Bath/media/Ethical`

Machine Learning: Algorithms and Ethics, U Bath, March 2018

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Peaceful coexistence with **intelligent autonomous systems (IASs)**?

- ▶ appropriate forms of **machine-control**
- ▶ appropriate forms of **human-machine-interaction**

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Existing societal processes are based on:

- ▶ **rational argumentation & dialog**
- ▶ **explicit normative reasoning** (legal & ethical)

Deployment of IASs lacking such competencies? How wise is this?

Talk Outline

- A Motivation:** Explicit Ethical Reasoning
- B Technology:** Universal Reasoning in Higher-Order Logic (HOL)
- C Evidence:** Analysis of Rational Arguments in Metaphysics
- E Demo:** Normative Reasoning Experimentation Platform

Long-term: Emerging Superintelligence

Really? Anyhow ...

- ▶ How to prevent IASs from turning against humanity?

Medium-term: Development of pseudo-ethical skills in IASs

- ▶ What architectural design? What functionalities?
- ▶ How to implement, deploy and verify?
- ▶ Which norms? Which reasoning principles?

Different kinds of systems and approaches:

- ▶ [Moor, 2009]:
 - ethical impact agents (ethical consequences to actions)
 - implicit ethical agents (ethical reactions to given situations)
 - explicit ethical agents (reasoning with ethical theories/rules)
 - full ethical agents (conscious, intentional, free will)
- ▶ bottom-up vs. top-down
- ▶ [DoranEtAl., 2017]:
opaque — comprehensible — interpretable — explainable AI

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Pseudo-Ethical IAS (medium-term)

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Pseudo-Ethical IAS (medium-term)

Related Work

- ▶ Artificial Moral Agents
 - ▶ [Wallach&Allen, 2008]
- ▶ Ethical Governors
 - ▶ [ArkinEtAl., 2009, 2012]
 - ▶ [Dennis&Fisher, 2017]
- ▶ Ethical Deliberation in ART
 - ▶ [Dignum, 2017]
- ▶ Programming Machine Ethics
 - ▶ [Pereira&Saptawijaya, 2016]
- ▶ ...

Which Reasoning Formalisms?

“If we had it [a *characteristica universalis*], we should be able to reason in metaphysics and morals in much the same way as in geometry and analysis.”

(Leibniz, 1677)

Challenges for Explicit Ethical Reasoning Engines

- ▶ Dilemmas, conflicting theories
- ▶ Appropriate handling of notion of **obligation**
 - ▶ Contrary-to-duty (**CTD**) scenarios

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Standard CTD structure (Chisholm)

1. obligatory 'a'
 2. obligatory 'if *a* then not *b*'
 3. if 'not *a*' then obligatory '*b*'
 4. 'not *a*'
- (in a given situation)

Danger: Paradox/inconsistency — ex falso quodlibet!

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CTD example (X. Parent): EU General Data Protection Regulation (GDPR)

1. Personal data shall be processed lawfully. (Art. 5)
E.g., the data subject must have given consent to the processing. (Art. 6/1.a)
2. **Implicit:** The data shall be kept, for the agreed purposes, if processed lawfully.
3. If personal data has been processed unlawfully, the controller has the obligation to erase the personal data in question without delay. (Art. 17.d, right to be forgotten)
4. **Given situation:** Some personal data has been processed unlawfully.

Danger: Paradox/inconsistency — ex falso quodlibet!

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Deontic Logic

- ▶ Reasoning about obligations and permissions
- ▶ Used to analyse normative reasoning in law
- ▶ Two groups of approaches:
 - Possible worlds
 - ▶ standard deontic logic CTD: no
 - ▶ dyadic deontic logic CTD: yes
 - Norm-based semantics
 - ▶ input/output logic CTD: yes

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L. van der Torre

X. Parent

Normative Reasoning Experimentation Platform

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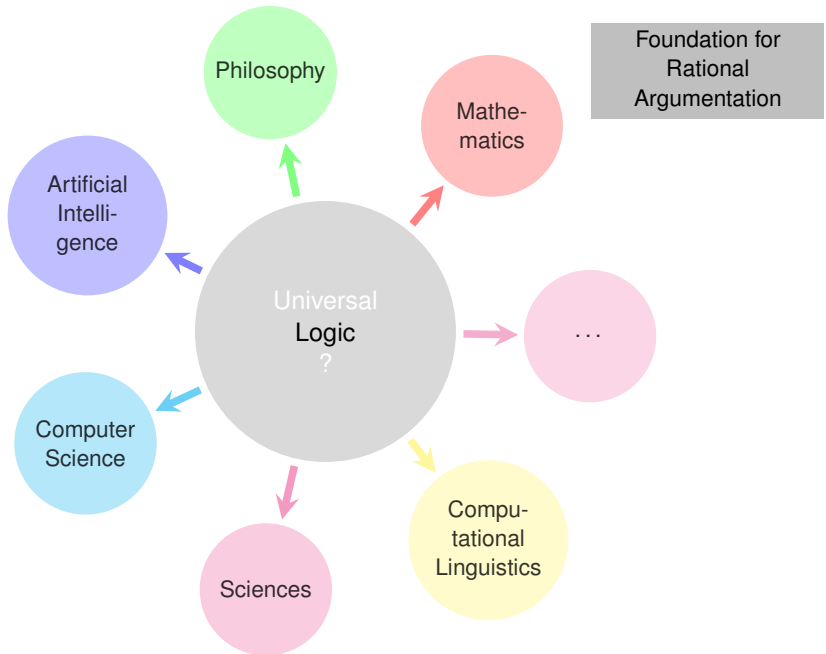
C. Benzmüller, 2018

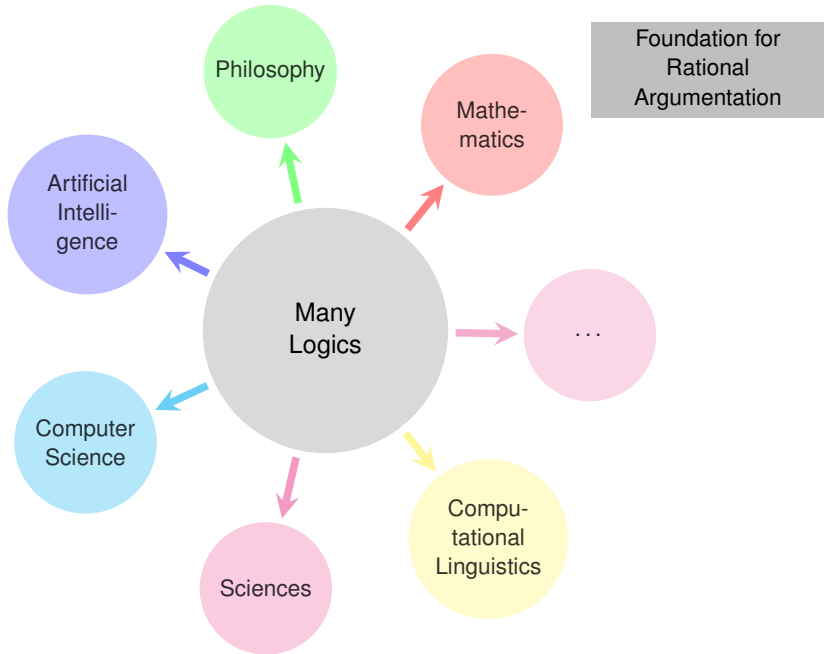
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Part B

Technology: Universal Reasoning in Higher-Order Logic







Logic Zoo

Classical Logic, of order

- 0. Propositional Logic
- 1. First-order Logic
- 2. Second-order Logic
- ...
- n. Higher-order Logic

Non-Classical Logics

- ▶ Intuitionistic/Constructive Logics (incl. Univalent Foundations)
- ▶ Modal Logics, Conditional Logics, Temporal Logics, Spatial Logics
- ▶ Many-valued Logics
- ▶ Paraconsistent Logics
- ▶ Free Logics, Inclusive Logics
- ▶ Logics for special applications: Ethics, Social Choice, Legal Reasoning, ...
- ▶ Separation Logic, ...

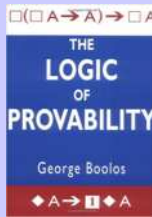
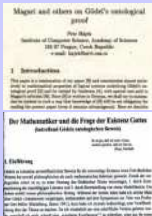
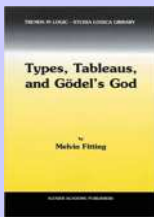
Example Application in Metaphysics/Philosophy:

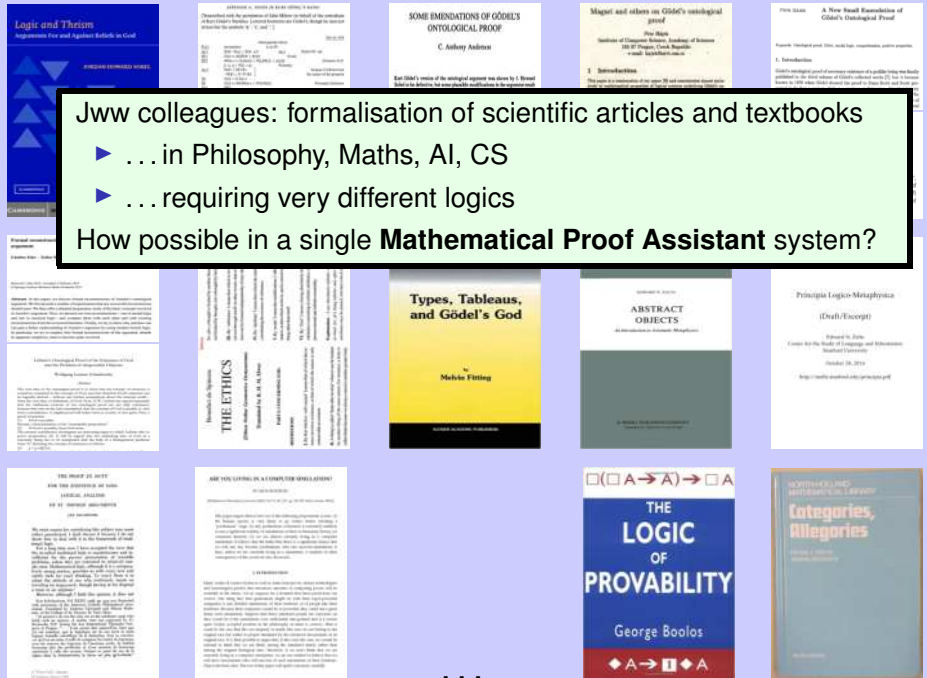
Necessarily, God exists:

Kurt Gödel's definition of God:

$$\Box \exists x. Gx$$

$$Gx := \forall \Phi. \text{Positive } \Phi \rightarrow \Phi x$$





Example: Modal Logic Textbook



STUDIES IN LOGIC AND PRACTICAL REASONING

VOLUME 3

D.M. GABBAY / P. GARDENFORS / J. SIEKMANN / J. VAN BENTHEM / M. VARDI / J. WOODS

EDITORS

Handbook of Modal Logic

2 BASIC MODAL LOGIC

In this section we introduce the basic modal language and its relational semantics. We define basic modal syntax, introduce models and frames, and give the satisfaction definition. We then draw the reader's attention to the internal perspective that modal languages offer on relational structure, and explain why models and frames should be thought of as graphs. Following this we give the standard translation. This enables us to convert any basic modal formula into a first-order formula with one free variable. The standard translation is a bridge between the modal and classical worlds, a bridge that underlies much of the work of this chapter.

2.1 First steps in relational semantics

Suppose we have a set of proposition symbols (whose elements we typically write as p, q, r and so on) and a set of modality symbols (whose elements we typically write as m, m', m'' , and so on). The choice of PROP and MOD is called the *signature* (or *similarity type*) of the language; in what follows we'll tacitly assume that PROP is denumerably infinite, and we'll often work with signatures in which MOD contains only a single element. Given a signature, we define the *basic modal language* (over the signature) as follows:

$$\varphi ::= p \mid \top \mid \perp \mid \neg\varphi \mid \varphi \wedge \psi \mid \varphi \vee \psi \mid \varphi \rightarrow \psi \mid \varphi \leftrightarrow \psi \mid \langle m \rangle \varphi \mid [m] \varphi.$$

That is, a basic modal formula is either a proposition symbol, a boolean constant, a boolean combination of basic modal formulas, or (most interesting of all) a formula prefixed by a diamond

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2.1 First steps in relational semantics

Metalanguage

What follows we tacitly assume that $PROP$ is countably infinite, and we'll often work with signatures in which MOD contains only a single element. Given a signature, we define the *basic modal language* (over the signature) as follows:

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Syntax

use elements we typically write as p, q, r and
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nature (or *similarity type*) of the language; in

Example: Modal Logic Textbook

A *model* (or *Kripke model*) $\mathfrak{M}$ for the basic modal language (over some fixed signature) is a triple $\mathfrak{M} = (W, \{R^m\}_{m \in \text{MOD}}, V)$. Here W , the *domain*, is a non-empty set, whose elements we usually call *points*, but which, for reasons which will soon be clear, are sometimes called *states*, *times*, *situations*, *worlds* and other things besides. Each R^m in a model is a binary relation on W , and V is a function (the valuation) that assigns to each proposition symbol p in PROP a subset $V(p)$ of W ; think of $V(p)$ as the set of points in $\mathfrak{M}$ where p is true. The first two components $(W, \{R^m\}_{m \in \text{MOD}})$ of $\mathfrak{M}$ are called the *frame* underlying the model. If there is only one relation in the model, we typically write (W, R) for its frame, and (W, R, V) for the model itself. We encourage the reader to think of Kripke models as graphs (or to be slightly more precise, *directed graphs*, that is, graphs whose points are linked by directed arrows) and will shortly give some examples which show why this is helpful.

Suppose w is a point in a model $\mathfrak{M} = (W, \{R^m\}_{m \in \text{MOD}}, V)$. Then we inductively define the notion of a formula φ being *satisfied* (or *true*) in $\mathfrak{M}$ at point w as follows (we omit some of the clauses for the booleans):

$\mathfrak{M}, w \models p$	iff	$w \in V(p)$,
$\mathfrak{M}, w \models \top$		always,
$\mathfrak{M}, w \models \perp$		never,
$\mathfrak{M}, w \models \neg\varphi$	iff	not $\mathfrak{M}, w \models \varphi$ (notation: $\mathfrak{M}, w \not\models \varphi$),
$\mathfrak{M}, w \models \varphi \wedge \psi$	iff	$\mathfrak{M}, w \models \varphi$ and $\mathfrak{M}, w \models \psi$,
$\mathfrak{M}, w \models \varphi \rightarrow \psi$	iff	$\mathfrak{M}, w \not\models \varphi$ or $\mathfrak{M}, w \models \psi$,
$\mathfrak{M}, w \models \langle m \rangle \varphi$	iff	for some $v \in W$ such that $R^m wv$ we have $\mathfrak{M}, v \models \varphi$,
$\mathfrak{M}, w \models [m] \varphi$	iff	for all $v \in W$ such that $R^m wv$ we have $\mathfrak{M}, v \models \varphi$.

Example: Modal Logic Textbook

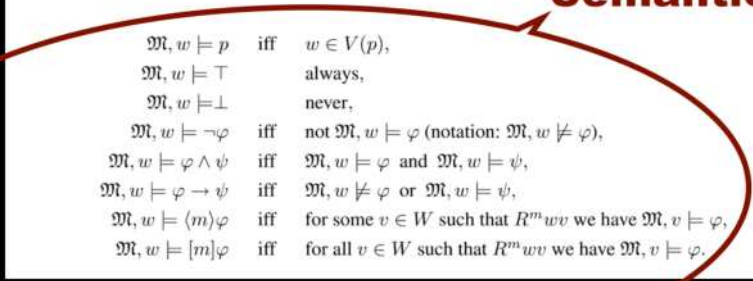
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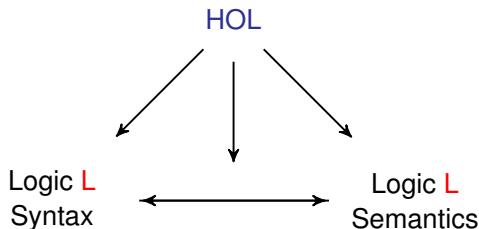
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Universal Reasoning in Meta-Logic HOL



Examples for **L we have already studied:**

Intuitionistic Logics, (Mathematical) Fuzzy Logics, Free Logic, Modal Logics, Description Logics, Conditional Logics, Access Control Logics, Hybrid Logics, Multivalued Logics, Logics based on Neighborhood Semantics, Paraconsistent Logics, Dyadic Logic, ...

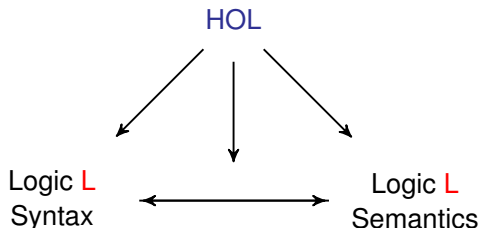
Embedding works also for quantifiers (first-order & higher-order)

HOL provers become universal logic reasoning engines!

interactive: Isabelle/HOL, PVS, HOL4, Hol Light, Coq/HOL, ...

automated: Leo-III, LEO-II, Satallax, TPS, Nitpick, Isabelle/HOL, ...

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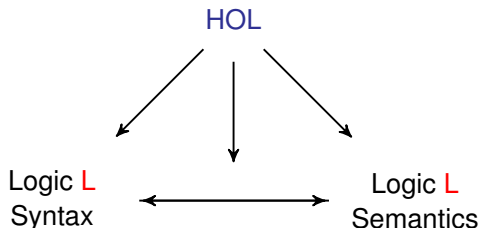
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Isabelle/HOL (one of various Theorem Provers for HOL)



Isabelle

**UNIVERSITY OF
CAMBRIDGE**
Computer Laboratory

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TECHNISCHE UNIVERSITÄT
MÜNCHEN

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What is Isabelle?

Isabelle is a generic proof assistant. It allows mathematical formulas to be expressed in a formal language and provides tools for proving those formulas in a logical calculus. Isabelle was originally developed at the [University of Cambridge](#) and [Technische Universität München](#), but now includes numerous contributions from institutions and individuals worldwide. See the [Isabelle overview](#) for a brief introduction.

Now available: Isabelle2017 (October 2017)



Download for
Mac OS X



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Some notable changes:

- Experimental support for Visual Studio Code as alternative PIDE front-end.
- Improved Isabelle/Edt Prover IDE: management of session sources independently of editor buffers, removal of unused theories, explicit indication of theory status, more careful auto-indentation.
- Session-qualified theory imports.
- Code generator improvements: support for statically embedded computations.
- Numerous HOL library improvements.
- More material in HOL-Algebra, HOL-Computational_Algebra and HOL-Analysis (ported from HOL-Light).
- Improved Nunchaku model finder, now in main HOL.
- SQL database support in Isabelle/Scala.

See also the cumulative [NEWS](#).

Distribution & Support

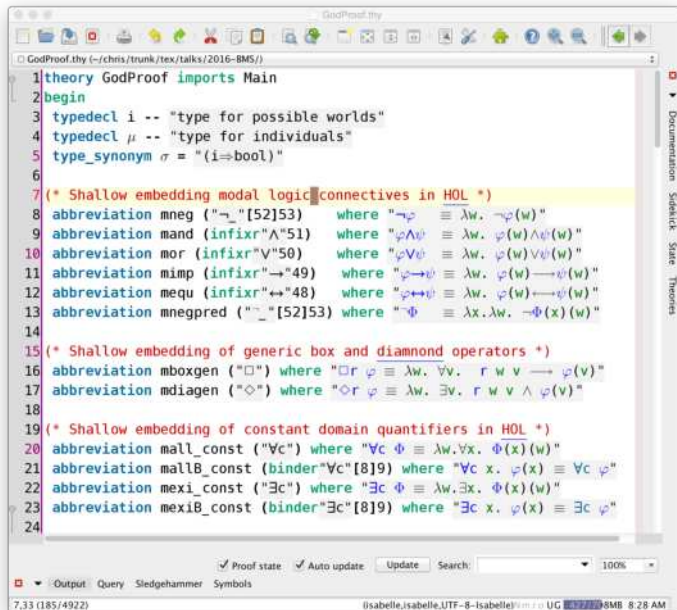
Isabelle is distributed for free under a conglomerate of open-source licenses, but the main code-base is subject to BSD-style regulations. The application bundles include source and binary packages and documentation, see the detailed [installation instructions](#). A vast collection of Isabelle examples and applications is available from the [Archive of Formal Proofs](#).

<https://isabelle.in.tum.de>

many other systems:

Coq, HOL, HOL Light, OVS, Lean, NuPrL, IMPS, ACL2, **Leo-II/Leo-III**, ...

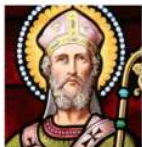
Universal Reasoning in Isabelle/HOL



```
1 theory GodProof imports Main
2 begin
3   typedcl i -- "type for possible worlds"
4   typedcl μ -- "type for individuals"
5   type_synonym σ = "(i⇒bool)"
6
7   (* Shallow embedding modal logic connectives in HOL *)
8   abbreviation mneg ("¬"[52]53) where "¬φ ≡ λw. ¬φ(w)"
9   abbreviation mand (infixr"∧"51) where "φ∧ψ ≡ λw. φ(w)∧ψ(w)"
10  abbreviation mor (infixr"∨"50) where "φ∨ψ ≡ λw. φ(w)∨ψ(w)"
11  abbreviation mimp (infixr"→"49) where "φ→ψ ≡ λw. φ(w)→ψ(w)"
12  abbreviation mequ (infixr"↔"48) where "φ↔ψ ≡ λw. φ(w)↔ψ(w)"
13  abbreviation mnegpred ("¬"[52]53) where "¬Φ ≡ λx.λw. ¬Φ(x)(w)"
14
15  (* Shallow embedding of generic box and diamond operators *)
16  abbreviation mboxgen ("□") where "□φ ≡ λw. ∀v. r w v → φ(v)"
17  abbreviation mdiagen ("◇") where "◇φ ≡ λw. ∃v. r w v ∧ φ(v)"
18
19  (* Shallow embedding of constant domain quantifiers in HOL *)
20  abbreviation mall_const ("∀c") where "∀c Φ ≡ λw.∀x. Φ(x)(w)"
21  abbreviation mallB_const (binder"∀c"[8]9) where "∀c x. φ(x) ≡ ∀c φ"
22  abbreviation mexi_const ("∃c") where "∃c Φ ≡ λw.∃x. Φ(x)(w)"
23  abbreviation mexiB_const (binder"∃c"[8]9) where "∃c x. φ(x) ≡ ∃c φ"
24
```

Ontological Proofs of God's Existence

A Long and Continuing Tradition in Philosophy



St. Anselm



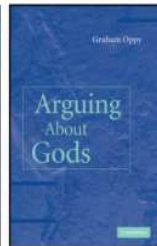
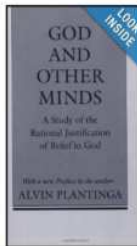
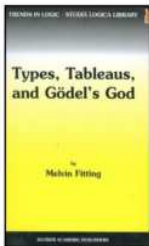
Descartes



Leibniz



Gödel



Computational Metaphysics: Kurt Gödel's Ontological Argument



Ontologische Beweise

Feb 10, 1970

$P(\varphi)$ φ is positive (i.e. $\varphi \in P$)

Ax 1 $P(\varphi) \rightarrow P(\psi) \supset P(\varphi \wedge \psi)$ Ax 2 $P(\varphi) \rightarrow P(\neg \neg \varphi)$

[1] $G(x) \equiv (\varphi) [P(\varphi) \supset \varphi(x)]$ (Good)

[2] $\varphi \text{ Ess } x \equiv (\psi) [\psi(x) \supset N(\exists y (\varphi(y) \supset \psi(y)))]$ (Essence of x)

$P \supset Nq = N(p \supset q)$ Necessity

Ax 2 $P(\varphi) \supset NP(\varphi)$
 $\neg P(\varphi) \supset N \neg P(\varphi)$ } because it follows from the nature of the property

Th. $G(x) \supset G \text{ Ess } x$

Df $E(x) \equiv (\varphi) [\varphi \text{ Ess } x \supset N \exists x \varphi(x)]$ necessary Existence

Ax 3 $P(E)$

Th. $G(x) \supset N(\exists y) G(y)$

hence $(\exists x) G(x) \supset N(\exists y) G(y)$

" $M(\exists x) G(x) \supset M N(\exists y) G(y)$

" $\supset N(\exists y) G(y)$

M = possibility

any two instances of x are nec. equivalent

exclusive or * and for any number of summands

$M(\exists x) G(x)$ means all possible (the system) This is:

Ax 4: $P(\varphi) \cdot \varphi \supset N \varphi$

hence $\{x = x\}$ is possible
 $\{x \neq x\}$ is not possible

But if a system S of axioms would mean that the axiom prop. A (which is positive) would be $x \neq x$

Positive means positive in the modal sense (independently of the accidental structure of the world). Only in the actual world. It also means "attribution" as opposed to "privation" (or containing privation). This requires a possible world.

Df φ positive: $(x) N \neg \varphi(x) \equiv \varphi(x) \supset N \neg \varphi(x)$

hence $x \neq x$ positive iff $x = x$ is necessary. At the exp. of pos. Ax 4

any x i.e. the formal form in terms of elem. prop. contains a member without negation.

Computational Metaphysics: Kurt Gödel's Ontological Argument

Ontologische Beweise

Feb 10, 1970

P(φ) φ is positive (i.e. $\varphi \in P$)

At 1 $P(\varphi) \rightarrow P(\neg \neg \varphi)$ At 2 $P(\varphi) \rightarrow P(\neg \neg \varphi)$

[1] $G(x) \equiv (\varphi) [P(\varphi) \rightarrow \varphi(x)]$ (Good)

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$M(\exists x) G(x) \supset M N(\exists y) G(y)$

$\supset N(\exists y) G(y)$

$M = possibility$

any two instances of x are nec. equivalent

exclusive or * and for any number of summands

$M(\exists x) G(x)$ means ^(the system of) all pos. props. is compatible
 This is true because of:

At 4 $P(\varphi) \cdot \varphi \supset N \neg \varphi \supset P(\neg \neg \varphi)$ which implies

$\begin{cases} x = x & \text{is positive} \\ x \neq x & \text{is negative} \end{cases}$

But if a system S of pos. props. were inconsistent it would mean that the sum prop. A (which is positive) would be $x \neq x$

Positive means positive in the modal sense (independently of the accidental structure of the world). Only in the actual time. It also means "attribution" as opposed to "privation" (or containing privation) - This requires a further proof

Df φ positive iff: $(x) N \neg (\varphi(x) \supset \varphi(x))$ $\varphi(x) \supset N \neg \varphi(x)$

hence $x \neq x$ positive iff $x = x$ is necessary

or the exp. of pos. At 4

x i.e. the formal form in terms of elem. prop. contains a member without negation.

Computational Metaphysics: Dana Scott's Variant

Axiom A1 Either a property or its negation is positive, but not both: $\forall\phi[P(\neg\phi) \leftrightarrow \neg P(\phi)]$

Axiom A2 A property necessarily implied by a positive property is positive:
 $\forall\phi\forall\psi[(P(\phi) \wedge \Box\forall x[\phi(x) \rightarrow \psi(x)]) \rightarrow P(\psi)]$

Thm. T1 Positive properties are possibly exemplified: $\forall\phi[P(\phi) \rightarrow \Diamond\exists x\phi(x)]$

Def. D1 A *God-like* being possesses all positive properties: $G(x) \leftrightarrow \forall\phi[P(\phi) \rightarrow \phi(x)]$

Axiom A3 The property of being God-like is positive: $P(G)$

Cor. C Possibly, God exists: $\Diamond\exists xG(x)$

Axiom A4 Positive properties are necessarily positive: $\forall\phi[P(\phi) \rightarrow \Box P(\phi)]$

Def. D2 An *essence* of an individual is a property possessed by it and necessarily implying any of its properties:
 $\phi \text{ ess. } x \leftrightarrow \phi(x) \wedge \forall\psi(\psi(x) \rightarrow \Box\forall y(\phi(y) \rightarrow \psi(y)))$

Thm. T2 Being God-like is an essence of any God-like being: $\forall x[G(x) \rightarrow G \text{ ess. } x]$

Def. D3 *Necessary existence* of an individual is the necessary exemplification of all its essences:
 $NE(x) \leftrightarrow \forall\phi[\phi \text{ ess. } x \rightarrow \Box\exists y\phi(y)]$

Axiom A5 Necessary existence is a positive property: $P(NE)$

Thm. T3 Necessarily, God exists: $\Box\exists xG(x)$



Computational Metaphysics: Scott's Variant

Axiom A1 Either a property or its negation is positive, but not both: $\forall \phi [P(\neg \phi) \leftrightarrow \neg P(\phi)]$

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Difference to Gödel (who omits this conjunct)

Computational Metaphysics: Vision of Leibniz (1646–1716) — *Calculemus!*



If controversies were to arise, there would be no more need of disputation between two philosophers than between two accountants. For it would suffice to take their pencils in their hands, to sit down to their slates, and to say to each other . . . : Let us calculate.

(Translation by Russell)

Quo facto, quando orientur controversiae, non magis disputatione opus erit inter duos philosophos, quam inter duos Computistas. Sufficiet enim calamos in manus sumere sedereque ad abacos, et sibi mutuo . . . dicere: calculemus. (Leibniz, 1684)



Required:
characteristica universalis and **calculus ratiocinator**

Computational Metaphysics: Scott's and Gödel's Variants — Demo

Axiom A1 Either a property or its negation is positive, but not both: $\forall \phi [P(\neg \phi) \leftrightarrow \neg P(\phi)]$

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$$\forall \phi \forall \psi [(P(\phi) \wedge \Box \forall x [\phi(x) \rightarrow \psi(x)]) \rightarrow P(\psi)]$$

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Computational Metaphysics: Scott's and Gödel's Variants — Demo

Axiom A1

$$\forall \phi [P(\neg \phi) \leftrightarrow \neg P(\phi)]$$

Axiom A2

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Def. D1

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Axiom A3

$$P(G)$$

Cor. C

$$\Diamond \exists x G(x)$$

Axiom A4

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Def. D2

$$\phi \text{ ess. } x \leftrightarrow \phi(x) \wedge \forall \psi (\psi(x) \rightarrow \Box \forall y (\phi(y) \rightarrow \psi(y)))$$

Thm. T2

$$\forall x [G(x) \rightarrow G \text{ ess. } x]$$

Def. D3

$$NE(x) \leftrightarrow \forall \phi [\phi \text{ ess. } x \rightarrow \Box \exists y \phi(y)]$$

Axiom A5

$$P(NE)$$

Thm. T3

$$\Box \exists x G(x)$$

Computational Metaphysics: Scott's and Gödel's Variants — Demo

Axiom A1

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Axiom A3

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Axiom A4

$$\forall \phi [P(\phi) \rightarrow \Box P(\phi)]$$

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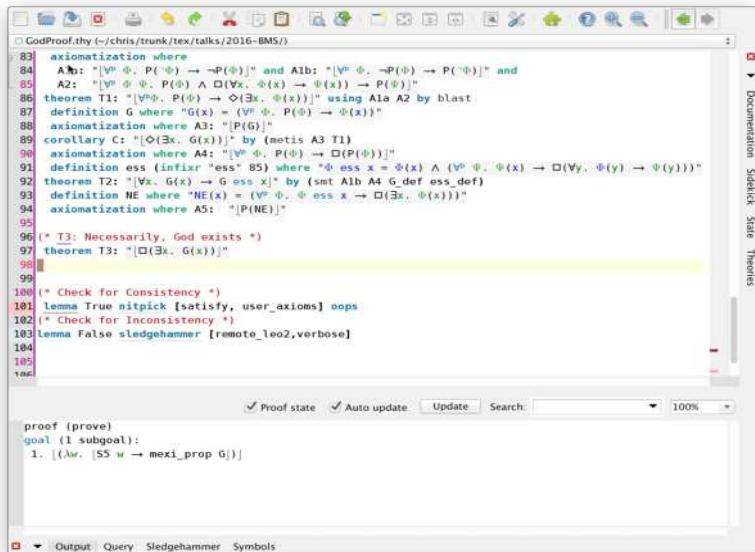
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Thm. T3

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Computational Metaphysics: Scott's and Gödel's Variants — Demo



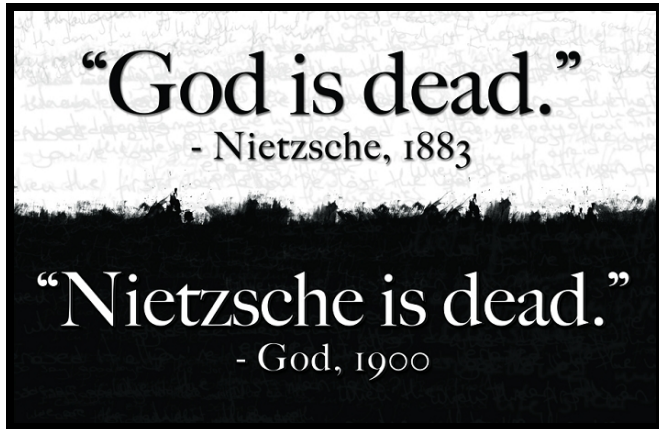
The screenshot shows the GodProof theorem prover interface. The main window displays a formal proof script in a text editor. The script defines axioms A1 and A2, a theorem T1, a definition G, and a corollary C. It then proves T2 and NE, and finally proves T3: "Necessarily, God exists". The script uses various logical connectives and quantifiers, and includes comments for consistency and inconsistency checks.

```
83 axiomatization where
84 A1: "[ $\forall \phi. P(\neg \phi) \rightarrow \neg P(\phi)$ ]" and A1b: "[ $\forall \phi. \neg P(\phi) \rightarrow P(\neg \phi)$ ]" and
85 A2: "[ $\forall \phi \psi. P(\phi) \wedge \Box(\forall x. \phi(x) \rightarrow \psi(x)) \rightarrow P(\psi)$ ]"
86 theorem T1: "[ $\forall \phi. P(\phi) \rightarrow \Box(\exists x. \phi(x))$ ]" using A1a A2 by blast
87 definition G where "G(x) = ( $\forall \phi. P(\phi) \rightarrow \phi(x)$ )"
88 axiomatization where A3: "[P(G)]"
89 corollary C: "[ $\Box(\exists x. G(x))$ ]" by (metis A3 T1)
90 axiomatization where A4: "[ $\forall \phi. P(\phi) \rightarrow \Box(P(\phi))$ ]"
91 definition ess (infix "ess" 85) where " $\phi$  ess x =  $\phi(x) \wedge (\forall \psi. \psi(x) \rightarrow \Box(\forall y. \psi(y) \rightarrow \psi(y)))$ "
92 theorem T2: "[ $\forall x. G(x) \rightarrow G$  ess x]" by (smt A1b A4 G_def ess_def)
93 definition NE where "NE(x) = ( $\forall \phi. \phi$  ess x  $\rightarrow \Box(\exists x. \phi(x))$ )"
94 axiomatization where A5: "[P(NE)]"
95
96 (* T3: Necessarily, God exists *)
97 theorem T3: "[ $\Box(\exists x. G(x))$ ]"
98
99
100 (* Check for Consistency *)
101 lemma True nitpick [satisfy, user_axioms] oops
102 (* Check for Inconsistency *)
103 lemma False sledgehammer [remote_leo2, verbose]
104
105
106
```

At the bottom of the window, there is a control bar with checkboxes for "Proof state" and "Auto update", an "Update" button, a search field, and a progress indicator set to 100%. Below this is a text area for the proof goal, which contains:

```
proof (prove)
goal (1 subgoal):
1. [( $\lambda w. [55 w \rightarrow mexi\_prop G]$ )]
```

The bottom status bar shows the current mode is "Output", with other options being "Query", "Sledgehammer", and "Symbols".



Part C

Evidence: Analysing Rational Arguments in Metaphysics

[BenzmüllerWoltzenlogelPaleo, ECAI, 2014 + IJCAI, 2016 + KI 2016 + ...]



Variant of Dana Scott

- ▶ premises are **consistent**
- ▶ all argument steps are **logically correct**
 - in logic **S5**
 - already in logic **KB**

Intermediate Conclusion:

With our technology...

... it is possible to verify (selected) masterpiece arguments in philosophy.



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- ▶ premises are inconsistent/contradictory
- ▶ everything follows!
- ▶ Philosophers had not seen this
- ▶ ... but my theorem prover LEO-II did

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Results of our Analysis

... we continue with Scott's version

Further corollaries we can prove

- ▶ Monotheism
- ▶ God is flawless (has only positive properties)
- ▶ ...
- ▶ Modal Collapse: $\varphi \rightarrow \Box \varphi$

- ▶ there are no contingent truths
- ▶ no alternative worlds
- ▶ everything is determined
- ▶ no free will



Challenge: Can the Modal Collapse be avoided (with minimal changes)?

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Can the Modal Collapse be avoided?

Kurt Gödel's version of the ontological argument was shown by J. Howard Sobel to be defective, but some plausible modifications in the argument result in a version which is immune to Sobel's objection. A definition is suggested which permits the proof of some of Gödel's axioms.

Magari and others on Gödel's ontological proof

Petr Hájek

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182 07 Prague, Czech Republic

e-mail: hajek@uivt.cas.cz

1 Introduction

This paper is a continuation of my paper [H] and concentrates almost exclusively to mathematical properties of logical systems underlying Gödel's ontological proof [G] and its variant by Anderson [A], with special care paid to Magari's criticism [M]. Since [H] is written in German, we shall try to summarise its content in such a way that knowledge of [H] will be not obligatory for reading the present paper (even it remains advantageous). Here we describe

Der Mathematiker und die Frage der Existenz Gottes (betreffend Gödels ontologisches Beweis)

Es ist gut, daß wir nicht wissen,
was wir glauben, und es fast ist,
(Gott, Schelling)

1. Einführung

Gödel in seinem wissenschaftlichen Bereich für die notwendige Existenz eines Gott-ähnlichen Wesens hat sowohl philosophischen als auch mathematischen Interesse geweckt. Zweck der vorliegenden Arbeit ist es, zu einer Deutung des Gödel'schen Textes beizutragen, 1. durch Kommentierung der einschlägigen Literatur und 2. durch Bereitstellung von etwas Modelltheorie. Die Arbeit enthält keinen philosophischen Beitrag. Während der letzten Jahre habe ich starke Meinungen über Gödels Glaubwürdigkeit geäußert, insbesondere auf dem Symposium zur Frage von Professor Gottfried Schelling, Bonn 1991, doch habe ich niemals beabsichtigt, eine Veröffentlichung über das Thema zu machen. Da ich wiederholt um eine schriftliche Version gebeten wurde, entschloß ich mich, schnell eine „unreife Kurzfassung“¹ zu schreiben, eine für eine

Understanding Gödel's Ontological Argument

FRODE BJØRDAL

In 1970 Kurt Gödel, in a hand-written note entitled "Ontologischer Beweis", put forward an ontological argument for the existence of God, making use of second-order modal logical principles. Let the second-order formula $P(F)$ stand for "the property F is positive", and let "God" signify the property of being God-like. Gödel presupposes the following definitions:

Gödel's version of the modal ontological argument for the existence of God has been criticized by J. Howard Sobel [S] and modified by C. Anthony Anderson [A]. In the present paper we consider the extent to which Anderson's emendation is defeated by the type of objection first offered by the Monk Gaunilo to St. Anselm's original Ontological Argument. And we try to push the analysis of this Gödelian argument a bit further to bring it into closer agreement with the details of Gödel's own formulation. Finally, we indicate what seems to be the main weakness of this emendation of Gödel's attempted proof.

PETR HÁJEK

A New Small Emendation of Gödel's Ontological Proof

Keywords: Ontological proof, Gödel, modal logic, comprehension, positive properties.

1. Introduction

Gödel's ontological proof of necessary existence of a godlike being was finally published in the third volume of Gödel's collected works [7]; but it became known in 1970 when Gödel showed the proof to Dana Scott and Scott presented it (in fact a variant of it) at a seminar at Princeton. Detailed history is found in Adams' introductory remarks to the ontological proof in [7]. The proof uses modal logic and its analysis is an exciting exercise in systems of formal modal logic. Needless to say, formal modal logic has found several

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- contributed to clarification of controversy —
- revealed various flaws and issues —

Further Experiments



Ed Zalta (Stanford)

Principia Logico-Metaphysica

Hyperintensional higher-order modal logic

Inconsistency/Paradox detected



Daniel Kirchner
(Mathematics, FU Berlin)

Further Experiments



Ed Zalta (Stanford)

Principia Logico-Metaphysica

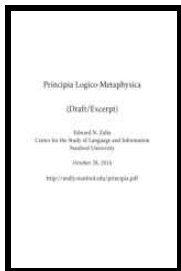
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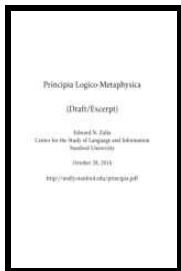
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Kirchner Paradox

Daniel & Isabelle/HOL have become close advisors of
Ed Zalta in the search for a repair

Computational Metaphysics par excellence!!!

Further Experiments



Ed Zalta (Stanford)

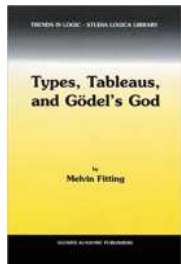
Principia Logico-Metaphysica

Hyperintensional higher-order modal logic

Inconsistency/Paradox detected



Daniel Kirchner
(Mathematics, FU Berlin)



Melvin Fitting (New York)

Ontological Argument (avoids modal collapse)

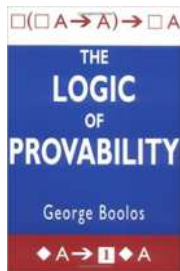
Intensional higher-order modal logic

Verified (main chapters)



David Fuenmayor
(Philosophy, FU Berlin)

Further Experiments



Textbook on Provability Logic

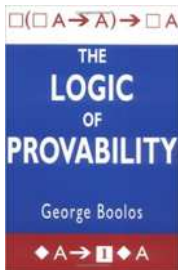
Provability Logic

Various parts verified



David Streit
(Mathematics, FU Berlin)

Further Experiments



Textbook on Provability Logic

Provability Logic

Various parts verified



David Streit
(Mathematics, FU Berlin)



Category Theory

Free first-order logic

(Constricted) Inconsistency detected



D. Scott
(UC Berkeley)

“If we had it [a *characteristica universalis*], we should be able to reason in metaphysics and morals in much the same way as in geometry and analysis.”

(Leibniz, 1677)

Part E

Demo: Normative Reasoning Experimentation Platform

Normative Reasoning Experimentation Platform

Normative Reasoning Experimentation Platform — Demo

`$HOME/GITHUBS/DeonticCorner/presentation/2018-Bath/media/GDPRmo`

Argued for explicit ethical reasoning competencies in IASs

- ▶ development of normative reasoning experimentation platform
- ▶ utilising HOL as universal meta-logic
- ▶ practical evidence from previous work
- ▶ suitable also for teaching

Ongoing and further work

(long-term challenge: prevent “human-unfriendly superintelligence”)

- ▶ formalisation and mechanisation of foundational ethical theories

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Before I forget:

— A big thanks to —

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Various Collaborators:



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(ANU Canberra)



Alexander Steen
(FU Berlin)



Max Wisniewski
(FU Berlin)



Ed Zalta
(Stanford U.)



Dana Scott
(UC Berkeley)

Many further Collaborators and Students:

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